GRADE 2 · DEBUGGING & READING CODE

Which One Is Wrong?

One script misses the goal — find it

Name: .... Date: ....

★ **I can compare two scripts to a goal and find which one has a bug.**

The goal: meet on 3

Bit (green) and Pixel (purple) both start on the green flag. They are SUPPOSED to MEET on number 3. But one of the two scripts has a bug, so it does not reach 3. Read both scripts, run them, and figure out WHICH sprite's script is wrong.

Number path 0–5

0	1	2	3	4	5
Bit			★ goal		Pixel

Bit's script

⚑ when green flag clicked

forward 3

Pixel's script

⚑ when green flag clicked

back 1

1 Run both

Where does each land? Bit (home 0, forward 3): ____ Pixel (home 5, back 1): ____. Do BOTH reach the goal 3?

____

____

2 Which one is wrong?

Which sprite's script is WRONG — Bit's or Pixel's? Write it. How do you know? (Hint: one sprite already reaches 3.)

____

____

3 How far off?

The goal is 3. The wrong sprite landed on ____. That is ____ number(s) away from the goal.

____

Big idea

To find a bug in two scripts, compare each one to the GOAL. The script that already reaches the goal is correct. The one that misses is the buggy one — and that tells you where to look.

TEACHER GUIDE · GRADE 2 · DEBUGGING & READING CODE

Which One Is Wrong?

Quick facilitation notes & answer key

What this worksheet teaches

Students learn the key debugging move: compare EACH set of moves to the goal to find the bug. The goal is to meet on 3; Bit reaches it and Pixel does not, so Pixel's set is the buggy one. Find which is wrong before fixing it. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: two counters on a number line 0 to 5 (green Bit, purple Pixel) so children can trace both sets of moves and see which one misses the goal.

How to lead it (about 20 minutes)

1. Show them (you do). Check each character against the goal: the one that misses has the bug.
2. Do one together. Exercise 1 — Bit reaches 3, Pixel lands on 4.
3. Let them try. Students name the buggy set and find how far off it is (Ex 2–3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit: 0, forward 3 → 3. Pixel: 5, back 1 → 4. Bit reaches the goal 3; Pixel does not.

Exercise 2 — Pixel's set is wrong; we know because Bit already lands on the goal, so the bug is in the other set.

Exercise 3 — the goal is 3; Pixel landed on 4, which is 1 number away. (This sheet only LOCATES the bug; fixing comes next.)

If students get stuck — and ways to adjust

Watch for: assuming both are wrong. Only the one that misses the goal has the bug.

Make it easier: mark the goal on the line and see which counter does not land there.

Make it harder: ask how they know it is Pixel and not Bit.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, find and fix the wrong one, and explain the fix.

You'll know it worked when

The child names the buggy set and says how far off the goal it landed.